

Octal -- Guided Notes

Unit tp-1.4 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Octal is a base-_____ number system. It uses only the digits _____ through _____. The digits 8 and 9 do _____ (do/do not) exist in octal. Each octal digit represents exactly _____ binary bits.

Question 2

Convert the octal number 37 to decimal:

3 is in the 8^1 position = $3 \times$ _____ = _____

7 is in the 8^0 position = $7 \times$ _____ = _____

Total = _____

Question 3

To convert binary to octal, group binary digits into sets of _____ starting from the right. Convert 110101:

Left group: _____ = octal _____

Right group: _____ = octal _____

Result: _____

Question 4

In Linux, file permissions are displayed and set using octal. The permission value 755 means: owner has _____ (read/write/execute), group has _____ (read/execute only), and others have _____ (read/execute only). The octal digit 7 = binary _____ = rwx.

Question 5

Using octal, the chmod command sets file permissions on Linux. chmod _____ grants full permissions to owner, and read/execute only to group and others. chmod 600 means owner has _____ only, and group and others have _____ permissions.

Question 6

Compare the four number systems by filling in the equivalent values for decimal 10:

Binary: _____

Octal: _____

Decimal: 10

Hexadecimal: _____

Question 7

Octal was historically used in older computing systems because early computers used _____ -bit or _____ -bit groupings where octal mapped cleanly. Today, octal is less common than hex because modern systems are byte-oriented (8-bit), and 8 bits divides more cleanly into _____ hex digits than _____ octal digits.

Question 8 -- Real World Application

REAL WORLD: A student is setting permissions on a web server script in Linux and runs `chmod 777` on the file. A senior admin immediately tells them to change it. Explain what `chmod 777` means using octal-to-binary conversion, why it is a security risk, and what a safer permission value would be and why.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/tp-1-4-octal/tp-1-4-octal-answer-key.pdf
```